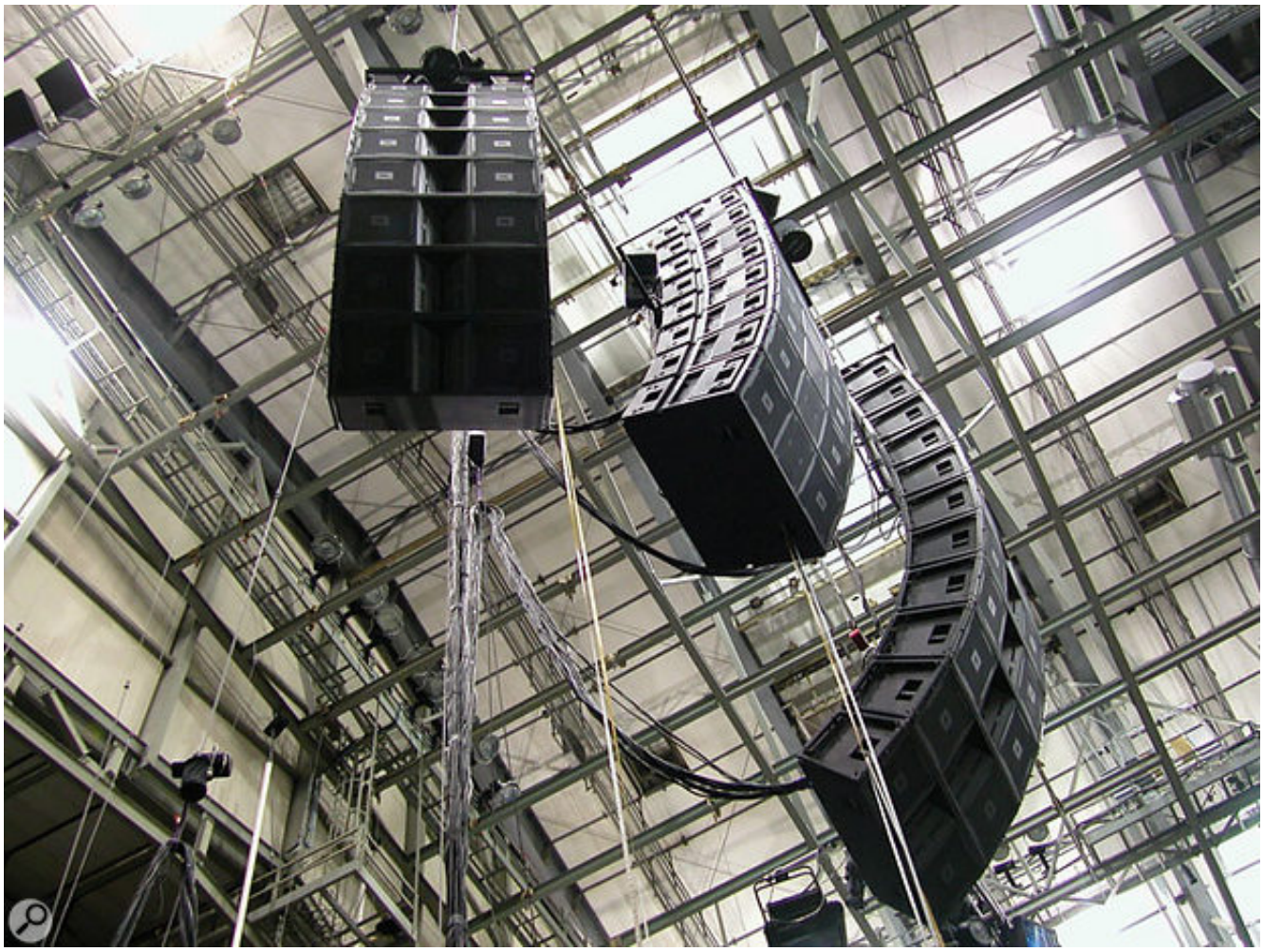
Line Arrays Explained

The Science And The Magic

Hardware > PA / Live Sound

By David Mellor



JBL Vertec line array. Photo courtesy of Harman Pro.

If you've been to a biggish gig or a festival in recent years, you've had the pleasure of hearing line arrays of loudspeakers in action. But why are line arrays the current 'best practice' in large–scale PA, how did they evolve, and will they ever filter down to more modest gig venues?

Here's a chance to show off what you know about live sound engineering. Simply complete the following sentence: The function of a PA system is to...

That wasn't hard, was it? But in case you're struggling, the function of a PA system is to deliver your sound to the audience, and deliver it well. It's as easy as that. But hang on, it doesn't seem to be all *that* easy, does it? Whenever have you experienced perfect sound as an audience member? And when have you ever felt that your band's sound has been delivered to the audience as well as it should have been? There must be additional criteria that need to be fulfilled to achieve satisfaction. And yes, there are. Three...

- Adequate level, in relation to purpose (clearly, heavy rock music needs to be louder than a classical guitarist).
- Low distortion, low noise and a flat frequency response.
- Adequate clarity, in relation to purpose (speech requires near–100 percent intelligibility; all the words in a theatre musical must be easily understood; other forms of music may not need to be absolutely crystal clear).

Achieving adequate level is never a problem. It hasn't been a problem since the 1970s, when PA systems as we know them today had fully matured. All you need is a recognition of how many watts you require for a particular venue, usually calculated by rule–of–thumb and reference to past experience, and the budget to hire enough amplifiers and loudspeakers. Achieving low distortion, low noise and a flat frequency response hasn't quite been fully solved, although if the noise level of your PA is audible to the audience there's a fault somewhere in the system: power amplifiers in general have a better signal–to–noise ratio than just about anything else you'll find in the whole of sound engineering. The frequency response of PA loudspeakers, however, leaves a lot to be desired, and it is definitely true to say that the only thing that produces more distortion than a loudspeaker is the lead guitarist's screaming Marshall on overdrive. But even though not all is yet perfect regarding the above points, most people find the sound quality of a decent PA system acceptable. And the typical sound of a PA has almost defined people's expectations of what a PA should sound like. A

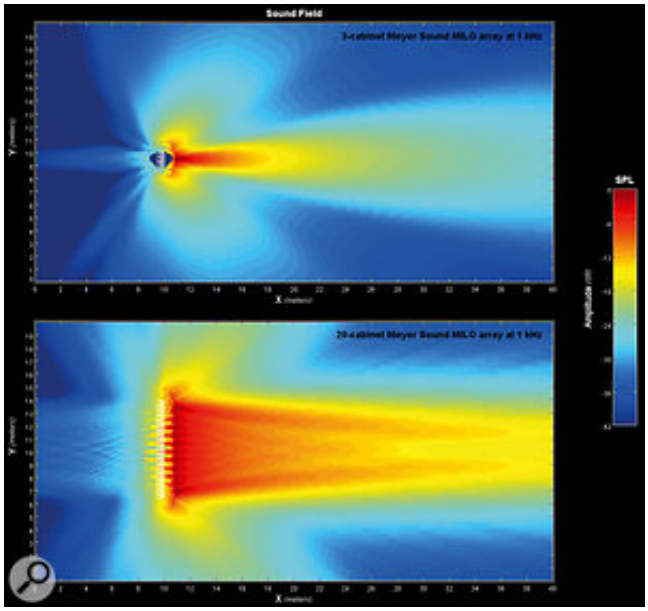
circular argument, perhaps, but there's a lot of truth in it.

There's still one point left unanswered: that of clarity. It is possible for a PA system to be capable of detailed, analytical clarity within itself. But when deployed in a real–life concert scenario it sounds anything but clear. You must have experienced it yourself many times as an audience member — that fuzzy mush of sound that clogs up your ears, but you can't really resolve it into music. Clarity, therefore, is the last unconquered frontier of PA. It is the last major problem that remains difficult to solve.

At this point I need to return to one of the requirements of PA that I previously said had been solved: that the PA system should be loud enough. There's no difficulty in making it loud enough, providing you have the budget — but it has to be loud enough for all members of the audience, and that's a problem that isn't necessarily solved just by spending a lot of money.

There are two scenarios here: one where the audience are seated, the other where they are standing and free to move. If the audience are free to move, it is acceptable to have different levels in different parts of the venue. Those who like it loud will gravitate towards the loudspeakers. Those who perhaps want to chat during the show will move further away. However, if the audience is entirely seated it suddenly becomes much more difficult. You don't want to deafen the front rows of the audience while leaving those at the back struggling to hear. If only certain members of the audience are delivered a level that is adequate, without being too quiet or too loud, the PA has not fully met its purpose. Let me therefore refine the requirements of PA into this simple statement: all of the audience should enjoy high–quality sound that is loud enough and clear enough.

Predicting Loudspeaker Behaviour With *MAPP*



MAPP Online is the *Multipurpose Acoustical Modeling Program* developed by loudspeaker manufacturer Meyer Sound to model the sound fields developed by its products in a variety of configurations. A sound designer is able to enter data into *MAPP Online*, including individual loudspeakers and arrays, then click the 'predict' button and get a graphical display of the expected coverage. Meyer Sound claim that *MAPP* will allow the user to:

- Plan an entire portable or fixed loudspeaker system and determine delay settings for fill loudspeakers.
- See interactions among loudspeakers and minimise destructive interference.
- Place microphones anywhere in the soundfield and predict the frequency response, impulse response and sound pressure at the microphone position.
- Refine system design to provide the best coverage of the intended audience area.
- Use a virtual equaliser to pre–determine the correct settings for best system response.
- Gain load information about the array, to determine rigging capacities.

Clearly, a system such as this, that is accurate in its predictions, is a tremendous tool for the sound designer. *MAPP Online* will run on Windows, Macintosh or Unix computers, and data that the user enters is sent to Meyer Sound's servers, where the analysis and prediction is made, then delivered back to your desktop. The example shown is a composite of two predictions made for arrays of Meyer Sound MILO cabinets, one with just three loudspeakers, the other with 20, both at 1kHz. It is clearly possible to

see how much more directional, and how much louder, the larger array is. You can also clearly see the 'side lobes' that develop — an unfortunate by–product of all line arrays that the sound designer must take into account.

Cover The Audience, Not The Walls

A paramount rule of PA is to direct the sound towards the audience and not elsewhere. But how often do you see this rule flouted? The best and most classic example of this *not* being done was in several London Underground stations, some years ago. At the time, the tube network was decaying and falling into disrepair, so several stations were refurbished with bright, modern designs. Along with the visual aspects, these stations were given new sound systems too. Some bright spark designer decided that the loudspeakers should be mounted in cylinders (cylinder = tube, get it?) and several should be mounted at intervals along the platform, parallel to the platform and just above waiting passengers' heads. The result was that from any point on the platform, you could hear every loudspeaker, with delays increasing with distance. It was, indeed, possible to stand as close to a speaker as you could and still not understand what was being said! This state of affairs wasn't allowed to continue for long, and now the speakers point as they should — down at the passengers on the platform.

So the most important thing is to point the loudspeakers at the people in the most direct way possible. At the same time, consider how much sound is being 'sprayed' onto the walls and ceiling. The audience will absorb much of the sound energy that strikes them, meaning that it won't be reflected to bounce around the auditorium and cause confusion. But the walls and ceiling are very likely to be reflective, so the more sound that goes in these directions, the more mush–inducing reflections will be created.

In a situation where the information content of speech is of primary importance, the classic solution to intelligibility is to use many small loudspeakers and have them close to the people — obviously, pointing at them and not at reflective surfaces. This works extremely well and the information content gets through clearly. But this solution is not acceptable for a musical performance. The reason for this is that we expect a performance to take place on a stage. We watch the performers on the stage, and we expect the sound to come from the stage too. If the sound were coming from a small speaker mounted at just a couple of metres distance, up and to the side, that would cause a conflict between the visual and the auditory. Everything might be clear and intelligible, but we wouldn't enjoy the performance.



The EAW CLA37 column loudspeaker uses seven 3–inch drive units to achieve a coverage of 120 degrees horizontal x 30 degrees vertical, thus controlling the vertical dispersion tightly. It is suitable for speech reinforcement in large reverberant environments if several or many units are distributed amongst the listeners.

So the multiple small speaker solution doesn't work for performance. We need the sound to seem as much as possible as though it comes from the stage, and for this you can't do better than actually having loudspeakers at the sides of the stage, like a great big stereo system. However, there are still potential problems...

The first problem has been mentioned already and has to do with directivity. Loudspeakers naturally have a characteristic directional response — almost omnidirectional at low frequencies, tightening to a focused beam at high frequencies. Put another way, anyone sitting directly in front of a loudspeaker will experience a reasonably flat frequency response, but people sitting further and further to the side will hear less and less high frequencies, so the sound will be increasingly dull. So the 'big stereo system' style of PA suffers in that it sprays the walls and ceiling with low–frequency and low–mid energy that reflects into a confusion of reverberation,

and only select members of the audience receive sound with a good balance of frequencies.

A second problem stems from the lack of directional control. Because much of the sound is spread widely, beyond the width of the audience, energy is lost. The more sound spreads out, the more thinly its energy is spread, and therefore the more level is lost with distance. This is an important point. The reason a sound source becomes apparently quieter as it becomes more distant is primarily because its energy is spread out. Yes, some level is lost through absorption in the air, but not much. It's distance that's the killer. An audience member sitting a long way from the loudspeakers will experience a distant and therefore quiet sound, while audience members close to the speakers are getting their heads blasted off!

Let's think in terms of light. Take a torch bulb. Intrinsically it emits light almost equally in all directions, so by itself it isn't much use for finding your way in the dark. But put a reflector behind it and a lens in front of it, so that its energy is concentrated into a beam, and you will notice immediately that it is now usefully bright. You'll also notice that the beam extends into the distance. So not only do you see the immediate area in front of your feet, but the area beyond where you direct the beam. The area of coverage is less, but you can now see where you're going. If the same could be done with loudspeakers, there would be two benefits: one, that the sound is focused on the audience and away from reflecting surfaces; and two, that the sound retains its level as it travels. So the audience members at the back are served as well as those at the front, and the difference in level between front and back is much less.

Directivity Theory

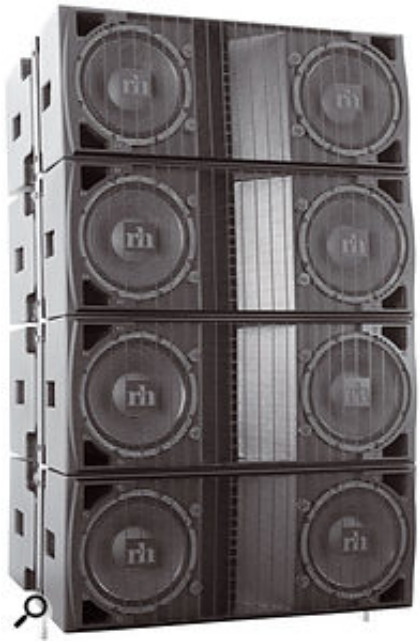
If you understand the theory behind the directional characteristics of sound sources, you'll be in a good position to understand PA loudspeakers and get the best out of them. There are two extremes of directionality, between

which there are other interesting cases. One extreme is the point source, which is a source of sound that has zero size. OK, there's no such thing as zero size, but in practice if a sound source is dimensionally smaller than the wavelength of sound it is emitting, it has the characteristics of a point source. The low–frequency output of a small loudspeaker would be a real–life example.

A point source emits sound equally in all directions. There you have it: all you need to know about the point source! Well, not quite all... but you'll need a little imagination. Imagine this very small point source pulsating outwards momentarily, just once. A sphere of high pressure leaves its surface and radiates outwards, becoming larger and larger. The point source has put a certain amount of energy into this pulse, and that same amount of energy over time has to cover a larger and larger area, the surface area of that continuously expanding sphere. I could at this point bore you to tears with detailed calculations concerning the surface area of a sphere, energy density and stuff like that, but instead I will cut directly to the chase and say this: for a point source, sound pressure decreases by 6dB for every doubling of distance. We call this the inverse square law.

One mistake or over–simplification is that it is commonly said that all sound obeys the inverse square law. This is not so. Only sound from a point source obeys the inverse square law. Any sound source that is not omnidirectional does not obey the inverse square law. (If you get so far away from it that visually it recedes to a point, from your point of observation it will appear to obey the inverse square law, but in practical terms this is not relevant to PA).

From this we can derive two interesting facts. The maximum rate at which sound level can decrease with distance is 6dB per doubling of distance. The only way sound can decay at a faster rate than that is if you actively do something to block it. Also, sound sources that are directional decay at a rate that is less than 6dB per doubling of distance.



A four–module array of Renkus–Heinz STLA/9 cabinets.

It's interesting to consider the opposite extreme. Would it be possible to have a sound source, the level from which does not decay at all with increasing distance? Amazingly, the answer is yes. It is possible to have a sound source that is so focused that it will cover an amazing distance with hardly any reduction in level. You want an example? I'll give you two examples: an old–fashioned ship's speaking tube, and a tin–can telephone. We call this kind of sound source a plane source. In both cases, the sound energy isn't just focused, it is constrained to travel within an enclosed medium so that it cannot spread out at all. And since it cannot spread, no level is lost. (In practice, a little level is lost, but nothing's perfect.) You can see that this is not a practical way of delivering your sound to the audience, so we will leave it as a curiosity, but a curiosity that demonstrates a useful principle.

The next type of sound source is the whole purpose of this article, and is the salvation of PA as we know it. We call this type of source — fanfare of trumpets — the line source. To understand it, lets go back to the point source for a moment. I said that the point source (which is omnidirectional) needs to be small in comparison with the wavelength of sound that it is emitting. The converse is true too: when a sound source is larger than the wavelength it is emitting, it becomes more directional. And the larger it is, the more tightly directional it is. So a really large sound source would be tightly directional. This is what we want: a source that can be focused and

directed to cover the audience, but not wasted on other areas of the auditorium.

But imagine you're a loudspeaker looking out from the stage to the audience. The audience in front of you are spread widely from left to right, but from top to bottom — in perspective, from the rear rows to the front — there is only a narrow spread. You can see the problem. If you made a large loudspeaker that focused the sound tightly enough to direct sound accurately in the vertical dimension, it wouldn't cover the full width of the audience. And vice versa: if it covered the full width, you'd end up covering the ceiling as well, and we know that's a bad thing.

The solution is to devise a loudspeaker that is tightly focused in the vertical dimension but spreads sound widely in the horizontal dimension. To do this, the speaker needs to be large vertically, but small horizontally. Like a column, in fact. And here we have it (bigger fanfare of trumpets): the column loudspeaker! Did I say 'column loudspeaker'? Sorry, I must use the more up–to–date and exciting terminology: line array. They are both examples of the line source.

Deploying The Line Array

Clearly, you're not going to have a full–scale line–array system in the back of your band's Transit van. In fact, playing through a line array for the first time may mark your transition from wannabe band to successful band. But there will come a time, hopefully, when you are called upon to have an influence in the specification of your touring PA system. At first, the line array looks intimidating. All those cabinets, all that cable. Who's going to go up there and string the whole thing together? The answer is nobody, because the system is assembled at stage level and the whole thing hoisted up. The motorised hoists even have remote controls so that no–one has to shout instructions or converse through an intercom. A line array can actually be set up by as few as two or three people. Any kind of flying, however,

involves considerable responsibility, and manufacturers are keen to use the words risk, damage, injury and death frequently in their operators' manuals. Apparently, the most dangerous part of the rigging process is when the equipment is at stage level. As it rises into the air, providing everything is done correctly and the equipment is in good condition, it flies out of the danger zone.

Setting up a conventional PA system on stage involves a certain amount of use of of rules–of–thumb. The line array is far too big a thing to set up in the same way, and once it's set up you don't really want to have to move it, so you need to be sure that the positioning is right, the height is right, the horizontal angling is right, and — most of all it — that the array takes up the optimum J–shaped curve to distribute sound evenly to the front and back of the audience, and everyone in between. To make this possible, manufacturers commonly provide software that can be used to calculate all the necessary parameters, examples of which are shown on the following two pages. Meyer Sound are good enough also to advise equipping yourself with binoculars, laser measuring tool, pedometer, laser inclinometer and a self–levelling, four–way laser. That should really be a last resort, as any decent venue should have a set of plans with accurate measurements!

The Column Loudspeaker

I often think that one of the best lessons of the past is not to go there again. However, the column loudspeaker has as important a place in the history of PA as the electric guitar does in rock music. Yes, really. One day there might be people who make a living as historians of PA, and they'll be able to tell us exactly how the column loudspeaker came to be developed. Until then, my guess is that it developed by chance and was found to work effectively. It seems like a natural development for a 1960s band to have speakers at either side of the stage for the vocals. Then they decide they want to be louder and need speakers with multiple drive units. But speakers that are wider take up more stage area, so they choose speakers that are taller. The

typical pub band of the 1960s would therefore have a pair of column loudspeakers, for vocals, that typically would contain four 10–inch or 12–inch drive units, sometimes topped off with a small horn (for example, the WEM Vendetta). Although they might seem primitive now, in fact they worked surprisingly well. The small horizontal dimension meant that the full width of the audience was covered, while the large vertical dimension ensured that the sound was 'beamed' to the back of the room. However, the next generation of bands working at a higher level of the business moved on to 'bins and horns'. (A horn loudspeaker is the most efficient way of converting amplifier power to sound. A 'bin' is a bass loudspeaker, which is commonly in the design of a folded horn. 'Bin and horn' systems of adequate physical size can sound very good, but their directionality is not necessarily well controlled.) Small bands followed suit with similar but scaled–down systems, and the column loudspeaker was forgotten. Small column loudspeakers, however, continued very successfully in speech PA, such as for places of worship, where intelligibility is all–important (see the photo below). The 'bin and horn' system amounted to nothing more than the 'big stereo' commented on earlier, and directional control was lacking.

The next real development in PA technology was the centre cluster, much used in musical theatre. The centre cluster relies on another directional technology known as the constant directivity horn. The idea here is to combine multiple full–range loudspeakers, each of which is designed to have a consistent directional pattern over a wide range of frequencies. Horn loudspeakers can be designed to do this reasonably well. These full–range loudspeakers are arrayed together into a part of a sphere and mounted high up to cover the whole of the audience. Each member of the audience is delivered sound through only one full–range loudspeaker (apart, of course, from people sitting exactly on the dividing line between the coverage of two loudspeakers).

The centre cluster is outstanding for its intelligibility. It fulfils the criterion of directing sound only at the audience, and has the additional benefit that it

forms a single sound source, therefore there is no possibility of hearing delayed sound from another loudspeaker somewhere else in the auditorium — at least, in a pure centre–cluster system. But there are two problems: the first is that ideally the centre cluster would be designed first, and then the auditorium designed around it! The second is that if each audience member is delivered sound (apart from the exception noted) by only one loudspeaker, plainly there is going to be a limit to how loud the sound can be. There will always be a role for centre clusters but, as we shall see, there are more flexible (literally) forms of loudspeaker distribution.

The Line Array

Although the column loudspeaker was effective in its context, it suffered from a lack of scale and a lack of science, each equally important. So to scale up a column loudspeaker to auditorium proportions took the best part of three decades. Still, we got there in the end. Here comes the science...

Going back to the point source, we find that level drops by 6dB for every doubling of distance. With the plane source, the level doesn't drop at all. So is there an in–between condition where the sound level drops by, say, 3dB? Yes there is, and it is the line source, which in theory can produce a cylindrical wave, as opposed to the spherical wave of the point source. A genuine cylindrical wave will have 360–degree dispersion in the horizontal dimension and zero dispersion in the vertical dimension. Any real–life source is going to be an approximation of this, but if someone offered you approximately £100, you would accept £75, wouldn't you?

Earlier, I said that to achieve directionality a sound source needs to be larger than the wavelength it is producing. To achieve focus, or near–zero dispersion, which is a more stringent requirement, it needs to be somewhere approaching four times the wavelength. The wavelengths of audible sound extend all the way to 17 metres (20Hz) and beyond. But taking a reasonable lowish frequency of 170Hz with a wavelength of two metres (taking 340 metres per second as a nice round figure for the speed of sound), a line source eight metres high will be necessary. Quite tall! But at least we have a notion with some science behind it.



A Meyer Sound 12–cabinet MICA array.

The next question is: how exactly do you make a loudspeaker that is several metres high? Currently, the way to do it is to stack multiple loudspeakers on top of each other. But instead of stacking 10–inch or 12–inch loudspeakers featuring identical drive units with poor HF response, as they did in the 1960s, each loudspeaker consists of LF and HF drive units and covers the full audio range (down to a reasonably low frequency). Also, rather than making one very tall cabinet, the modern line array consists of multiple small cabinets. The benefit of multiple cabinets is that you can assemble a line array that is as big or small as you like, or can fit in, or can budget for. You can also manipulate the shape of the array, which, as we shall see shortly, has significant benefits. Time for more science...

Since the line array is not actually one single tall–but–narrow drive unit, but is made up from discrete loudspeaker cabinets, one has to ask whether the individual units will couple together as though they were a genuine line source? The answer is yes, they will, but only where the drive units are separated by less than half a wavelength. This is easy for the lower frequencies, but more difficult to achieve as the wavelength shortens. As a benchmark, the wavelength at 400Hz is around 85 centimetres. So to couple at 400Hz the cabinets have to be less than 42.5 centimetres high.

OK, that's doable, but we are not even halfway up the audio band here.

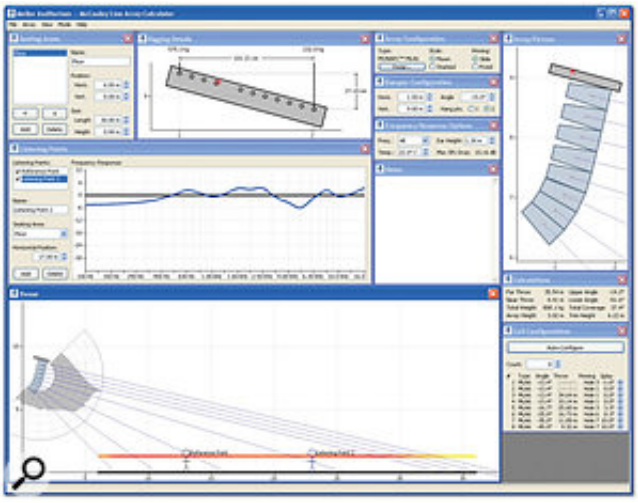
Still, at least we know the criteria to aim for. The longer the array is, the more tightly directional it will be in the vertical dimension, and for individual cabinets to couple well into the array, they have to be small vertically. The better both of these criteria can be achieved, the more controllable the beam of sound from the array will be. A good point is made by Ralph Heinz of PA manufacturers Renkus–Heinz: "The answer to the question of whether a line array is a line source is 'almost never'." Heinz's comment demonstrates that a theoretically perfect line source is virtually impossible to achieve. Only the best line arrays will come close.

Waveguide

I wouldn't be surprised if some of the readers out there are microwave engineers concerned with the efficient transmission and reception of microwave signals, *SOS* readers tending towards the technical. To you guys and girls, I'd like to say thanks — you gave us all the technology we need to make great–sounding line arrays. Seriously, a lot of loudspeaker technology does borrow from microwave technology, as the wavelengths of microwaves and sound waves are comparable. I have said already that to couple together into a line source, or at least a close approximation of a line source, individual sound sources must be no further apart than half a wavelength. You can turn this around and say that the closer together the individual sound sources are, the higher up the frequency spectrum line–source behaviour will be maintained. So each individual cabinet must be as short as possible in the vertical dimension. For preference, the height of the cabinet should be no more than the diameter of the low-frequency drive unit plus the thickness of the cabinet walls. However, to achieve a high sound level, clearly the low–frequency drive units will have to be reasonably large. In the Meyer Sound M3D, for example, 15–inch (38cm) drive units are employed on either side of the high–frequency unit. Since 38cm is half a wavelength at around 450Hz, an array of M3D cabinets will approximate to

a line source up to around this frequency. Above 450Hz, the directional characteristics will begin to depart from the ideal cylindrical wave, although not immediately.

So what happens above 450Hz in the case of the M3D? At 580Hz the signal is crossed over from the low–frequency drive units to a specially designed high–frequency driver. What is special about the design? Well, to make the whole concept of the line array viable, each individual cabinet has to be a line source in its own right, or at least approximate a line source as closely as possible. For this, the high–frequency drive unit needs very sophisticated design to produce the required wavefront that diverges hardly at all in the vertical dimension. There are several possible techniques for doing this — some practical, some not.



Many manufacturers of line array systems provide software for calculating optimum configuration and placement, which clearly will depend on each individual venue. Here we can see a calculator from McCauley that's particularly striking in its visualisation. You can enter the type of cabinet and quantity to be deployed, and the dimensions of the area to be covered. The software instantly shows the necessary angling of the cabinets (top right) with rigging information (top centre–left). In the lower–left panel, on the left we can see the line array surrounded by a graphic showing the vertical dispersion pattern. Listening points can be selected and the frequency response at those points will be displayed in the panel at centre left.

One possibility is the ribbon drive unit, which basically has a long, thin diaphragm up to around 15cm high. Incorporated into line–array cabinets, the ribbon driver will display at least reasonable line-source behaviour above around 4.5kHz, but below that point adjacent units will be more than half a wavelength apart and therefore will not couple correctly. For good coupling at higher frequencies, the high–frequency driver should radiate over at least 80 percent of the height of the cabinet. Ribbon drivers, in any case, are rather low on output when compared to more conventional compression drive units. A horn with

compression driver would be another possible choice, but for a horn to have a suitable direction pattern and a mouth area covering 80 percent of the height of a typical enclosure it would need to be inconveniently long. A reflector can also be used to focus sound, as in the Nexo GEO system. However, it seems that the current favourite technique is the acoustic lens.

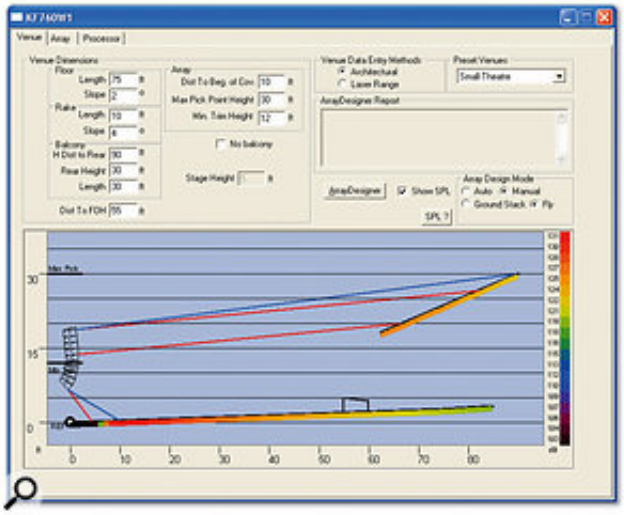
It's worth thinking for a moment about how an acoustic lens could be created. A lens for light works by slowing down light rays in a transparent medium of higher refractive index than air — i.e glass. This could be done for sound. Simply form a suitable medium into a lens shape and situate it in front of the drive unit. Sounds too simple to work? No, not at all, and this technique is indeed employed by Electro–Voice and McCauley. The lens is made out of foam, which acts as an 'obstacle array' around which the sound wave has to pass, thus slowing it down. The foam doesn't have to have the conventional lens shape, as it can be of variable density, which provides the 'shaping'. Foam does have its limitations, as you would expect. At high frequencies it will absorb sound rather than slow it down, and at low frequencies it will have no effect. Nevertheless, the fact that it is used for some current line–array systems demonstrates that it is a viable solution to the problem.

The other way of producing an acoustic lens is the path–length refractor. This uses metal plates to direct sound through channels. The channels have varying lengths and therefore sound can be slowed down by varying amounts of time. With appropriate design, this can form a perfectly viable lens that works over a reasonably wide range of frequencies. Obviously, since there are four different techniques currently in popular use in this application, the ultimate solution hasn't quite been found yet.

Intensity Shading & Divergence Shading

We've covered a lot of technical material so far, and it's worth going back for a moment to the purpose of the line array, which is to deliver sound to

the entire audience, at pretty much the same level, all the way from the front to the back. It does that by focusing the sound vertically while allowing it to spread out horizontally. Even though a well–designed line array can achieve that reasonably successfully, it will remain the case that the front of the audience receives a higher sound–pressure level than the rear of the audience — which, of course conflicts with our requirement. The solution to this is intuitive: simply reduce the output of the lower section of the array. This is known as intensity shading. The front rows of the audience are much closer to the lower cabinets than they are to the upper cabinets of the array, therefore reducing the level from the lower cabinets will deliver a lower sound–pressure level to the front of the audience. However, there is a problem here: the front rows will still hear sound coming from the upper cabinets of the array, and they will hear it clearly because these cabinets are louder. But sound from the higher cabinets will be delayed with respect to the lower cabinets, and that will create an interference pattern and an uneven distribution. This problem could be tackled with equalisation and delay, but that would destroy the elegant concept that the line array is.



This EAW calculator superficially doesn't look as polished as the McCauley one, but it offers presets for different types of venue. This example shows a small theatre with a balcony, which is also covered by the line array. The software shows, by colour coding, the levels that can be achieved in different sections of the venue.

The alternative to intensity shading is simple and obvious, and you would probably do it by instinct anyway. When a line array is flown, it will take you precisely two seconds to observe that the front rows of the audience are almost underneath the array, whereas the rear rows are much more on the same level as the top of the array. So it seems appropriate to curve the lower section of the array so that it points down at the front of the audience.

You have just created the familiar 'J' shape of the practical line array (see screen above). You have also implemented divergence shading. Simply by angling the cabinets apart more, you have required the sound they produce to cover a wider angle, therefore its intensity will be reduced at the listening position. Ideally this requires a more divergent cabinet for the curved part of the J, which manufacturers solve by designing specific long–throw and front–fill cabinets.

Line Arrays For The Gigging Band

If line arrays are good for top touring acts, surely they're good enough for the small gigging band too? In my view, it can only be a matter of time before manufacturers of small PA systems (many of whom make large–scale systems too) bring the line array into the small pub and club venue. There is a vacuum at the moment that desperately needs to be filled. Oddly enough, since a large–scale line array is composed of multiple small cabinets, there is absolutely no reason why you couldn't stand one on stage and stack it all the way to the ceiling, taking safety precautions of course. The limitation on small–scale deployment of line–arrays is actually ceiling height. In a large auditorium, the line array is hoisted high over the audience, so that the lower section of the J–shape points down at the front rows, while the upper cabinets point roughly horizontally at the rear of the audience. Raising the array like this reduces the difference in distance between front and rear, thus reducing the level difference due to distance. In a small venue, the line array would fire into the audience as much as it fires over their heads. Although the approximation to a cylindrical wave it produces would be advantageous, the loss of the downward perspective severely limits this advantage. And, of course, in a small venue the audience bunch up around the stage, so the front rows are very much closer to the loudspeakers than the people at the rear.

Contacts

- www.eaw.com
- www.electrovoice.com
- www.jblpro.com
- www.meyersound.com
- www.mccauley.com
- www.nexo.fr
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By now, you should be realising that there's a awful lot to know about line–array technology. There simply isn't room to cover it all here, plus the top manufacturers are constantly researching new developments, particularly with regard to focusing and steering of arrays. Although we wait for line–array technology to re–emerge as a major force at the gigging band level, my expectation is that it will. Although line arrays need to be large to work at their best, in respect of directional characteristics, there is no reason why smaller bands should not take advantage of the technology. Indeed JBL have scaled down that contained in their large–scale Vertec series (shown at the start of this article) into the new VRX932LA, designed for smaller venues. Each cabinet contains a 12–inch LF drive unit and an HF horn, designed for arrayability. Practical array sizes start at just two or three cabinets, and JBL advise up to six for optimum control over dispersion. A six–cabinet array would have to be flown, just like a full–scale line array, but JBL have cleverly provided the option of mounting two cabinets on a tripod stand, or on a pole on top of the SRX718S subwoofer.

An understanding of the directional properties and coverage of loudspeakers and arrays can only benefit the successful delivery of sound to the audience. Great sound should not only be the province of the large-scale auditorium PA, but should be available to all, at a reasonable price.

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